MATH 345 Skeleton Notes
Unit 4: Orthogonality and least squares
Section 4.3: Least squares as projection

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Revisiting a requested output as equations (U2-LO1, U2-LO3) asked whether a target vector $\mathbf{b}$ is reachable as $A\mathbf{x}$.
Recall the matrix and target from that activity:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we have

$$A\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}.$$

Every reachable output therefore has third coordinate equal to the sum of its first two coordinates. The target $\mathbf{b}$ is not reachable because $4 \neq 1 + 2$.
More generally, when $\mathbf{b} \notin \operatorname{col}(A)$, no input reaches the target exactly. Least squares replaces the unreachable target by the closest reachable output.

Definition: Least-squares solution, fitted vector, and residual

Let A be an $m \times n$ matrix and let $\mathbf{b} \in \mathbb{R}^m$.

A [ ] is a vector $\widehat{\mathbf{x}} \in \mathbb{R}^n$ that minimizes

$$\|A\mathbf{x} - \mathbf{b}\|^2$$

over all $\mathbf{x} \in \mathbb{R}^n$.
The vector

$$\widehat{\mathbf{b}} = A\widehat{\mathbf{x}}$$

is the [ ]. The vector

$$\mathbf{r} = \mathbf{b} - \widehat{\mathbf{b}} = \mathbf{b} - A\widehat{\mathbf{x}}$$

is the [ ].
The quantity

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^{m} \big((A\mathbf{x})_i - b_i\big)^2$$

is the sum of the squares of the coordinate-wise errors. This explains the name least squares. Minimizing the norm or its square gives the same minimizers.
If $\mathbf{b} \in \mathrm{col}(A)$, the minimum error is 0, and least squares agrees with solving $A\mathbf{x} = \mathbf{b}$ exactly. If the target is not reachable, the fitted vector $\widehat{\mathbf{b}}$ is the closest reachable output.
In A vector x is decomposed into its projection p along the subspace U and a perpendicular residual v. A right-angle marker shows that the residual is orthogonal to U, take the subspace to be $\mathrm{col}(A)$, relabel the original vector as $\mathbf{b}$, and relabel its projection as $\widehat{\mathbf{b}}$. The residual is

$$\mathbf{r} = \mathbf{b} - \widehat{\mathbf{b}}.$$

Projection theorem guarantees a closest vector $\widehat{\mathbf{b}} \in \mathrm{col}(A)$. Since $\widehat{\mathbf{b}}$ lies in the column space, at least one coefficient vector $\widehat{\mathbf{x}}$ satisfies

$$A\widehat{\mathbf{x}} = \widehat{\mathbf{b}}.$$

Therefore the least-squares fitted vector is

$$\widehat{\mathbf{b}} = A\widehat{\mathbf{x}} = \mathrm{proj}_{\mathrm{col}(A)}(\mathbf{b}).$$

Its residual

$$\mathbf{r} = \mathbf{b} - A\widehat{\mathbf{x}}$$

is orthogonal to $\mathrm{col}(A)$.

Fact: A^T checks column orthogonality
Write

$$A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}.$$

For any vector $\mathbf{r} \in \mathbb{R}^m$,

$$A^T\mathbf{r} = \begin{bmatrix} \mathbf{a}_1^T\mathbf{r} \\ \vdots \\ \mathbf{a}_n^T\mathbf{r} \end{bmatrix}.$$

Therefore

$$A^T\mathbf{r} = \mathbf{0}$$

exactly when $\mathbf{r}$ is orthogonal to every column of A. Equivalently,

$$\mathrm{col}(A)^\perp = \mathrm{null}(A^T).$$

Theorem: Least squares and the normal equations
A vector $\widehat{\mathbf{x}}$ is a least-squares solution exactly when the residual

$$\mathbf{r} = \mathbf{b} - A\widehat{\mathbf{x}}$$

is orthogonal to $\mathrm{col}(A)$. Equivalently,

$$A^T\left(\mathbf{b} - A\widehat{\mathbf{x}}\right) = \mathbf{0}.$$

Thus the least-squares solutions are exactly the solutions of

the [ ]

$$A^T A\widehat{\mathbf{x}} = A^T\mathbf{b}.$$

Why is this true?

Every candidate fitted vector $A\mathbf{x}$ lies in $\mathrm{col}(A)$. By Projection theorem, $A\widehat{\mathbf{x}}$ is closest to $\mathbf{b}$ exactly when

$$\mathbf{b} - A\widehat{\mathbf{x}} \in \mathrm{col}(A)^{\perp}.$$

By A^T checks column orthogonality, this is equivalent to

$$A^T\left(\mathbf{b} - A\widehat{\mathbf{x}}\right) = \mathbf{0}.$$

Expanding and rearranging gives

$$A^T\mathbf{b} - A^T A\widehat{\mathbf{x}} = \mathbf{0},$$

or

$$A^T A\widehat{\mathbf{x}} = A^T\mathbf{b}.$$

The fitted vector $\widehat{\mathbf{b}}$ is unique because orthogonal projection onto a subspace is unique. The coefficient vector $\widehat{\mathbf{x}}$ need not be unique. For any $\mathbf{z} \in \mathrm{null}(A)$,

$$A\left(\widehat{\mathbf{x}} + \mathbf{z}\right) = A\widehat{\mathbf{x}}.$$

When the columns of A are linearly independent, $\text{null}(A) = \{\mathbf{0}\}$ by the matrix characterization of linear independence, so the least-squares coefficient vector is unique in that case.

Fact: Independent columns and A^TA

Let A be an $m \times n$ matrix. Then A^TA is invertible if and only if the columns of A are linearly independent.

Why is this true?

If

$$A^TA\mathbf{x} = \mathbf{0},$$

then

$$0 = \mathbf{x}^TA^TA\mathbf{x} = \|A\mathbf{x}\|^2.$$

Therefore $A\mathbf{x} = \mathbf{0}$. If the columns of A are linearly independent, then

$$\text{null}(A) = \{\mathbf{0}\},$$

so $\mathbf{x} = \mathbf{0}$. Thus $\text{null}(A^TA) = \{\mathbf{0}\}$, and the square matrix A^TA is invertible. Conversely, suppose A^TA is invertible and

$$A\mathbf{x} = \mathbf{0}.$$

Multiplying by A^T gives

$$A^TA\mathbf{x} = \mathbf{0}.$$

Since A^TA is invertible, $\mathbf{x} = \mathbf{0}$. Therefore

$$\text{null}(A) = \{\mathbf{0}\},$$

so the columns of A are linearly independent.

Activity: Unit 2 target revisited (U4-LO4, U4-LO5)

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

1. Recall why $A\mathbf{x} = \mathbf{b}$ has no exact solution.
2. Form $A^T A$ and $A^T \mathbf{b}$.
3. Solve the normal equations for $\hat{\mathbf{x}}$.
4. Compute the fitted vector

   $$\hat{\mathbf{b}} = A\hat{\mathbf{x}}$$

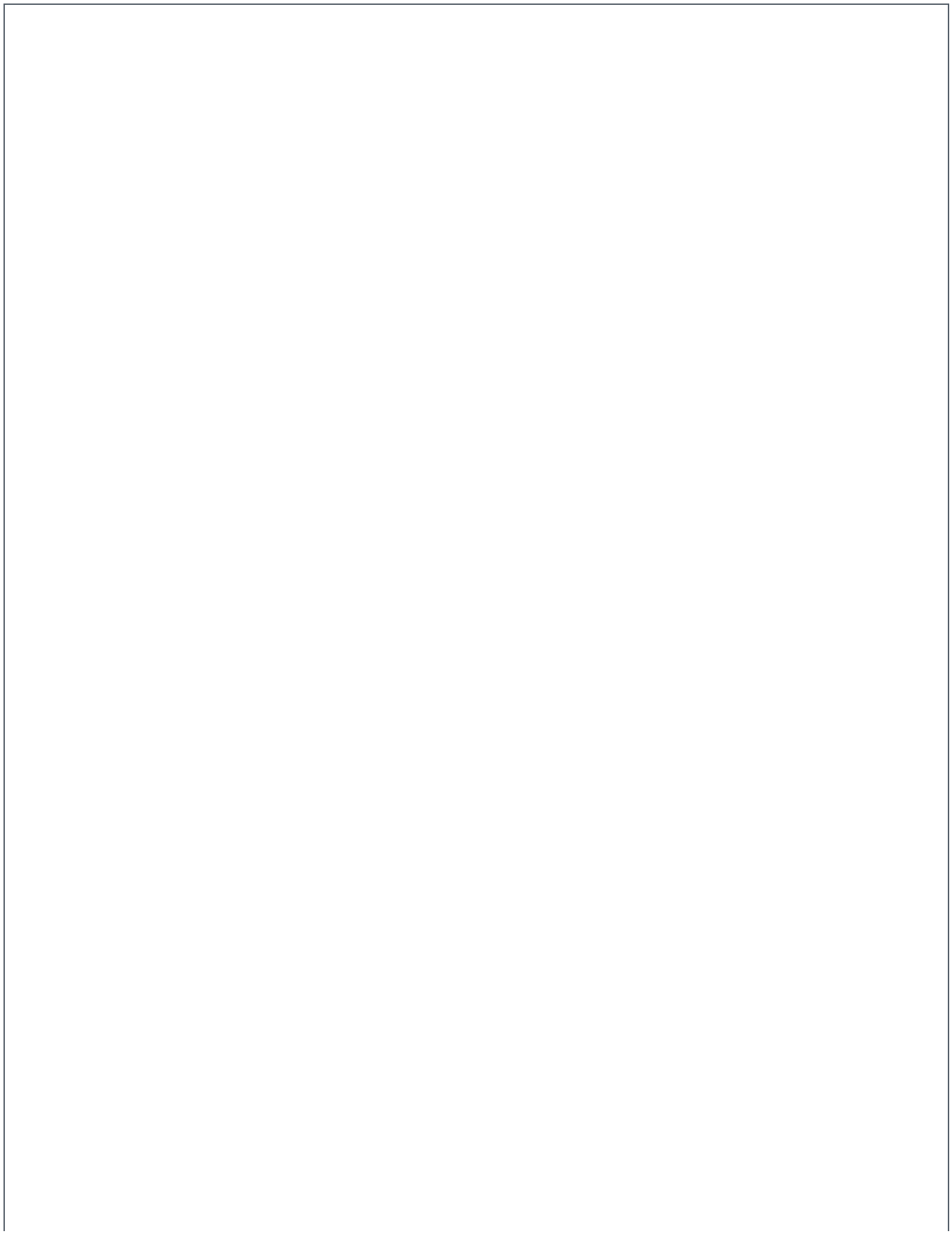
   and the residual

   $$\mathbf{r} = \mathbf{b} - \hat{\mathbf{b}}.$$

5. Check that

   $$A^T \mathbf{r} = \mathbf{0}.$$

6. Compute $\|\mathbf{r}\|^2$.
7. Which vector is the closest reachable output: $\hat{\mathbf{x}}$ or $\hat{\mathbf{b}}$?
8. The residual is not zero. Explain why this does not mean least squares failed.



Activity: Least squares and residual orthogonality (U4-LO4, U4-LO5)

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

1. Find the least-squares solution $\hat{\mathbf{x}}$ by solving the normal equations.
2. Compute the residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$ and check that it is orthogonal to every column of A.

Activity: Least squares with three unknowns (U4-LO4, U4-LO5)

Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}.$$

Find the least-squares solution $\widehat{\mathbf{x}}$.

Linear regression

An important practical application of least squares is

[ ], which fits a linear model to observed data to describe relationships and make predictions. Suppose data are given as pairs

$$(t_1, y_1), \ldots, (t_m, y_m).$$

A line model predicts

$$\hat{y} = c_0 + c_1 t.$$

The coefficient c_0 is the [ ], and c_1 is the [ ].

At input t_i, the predicted value is

$$\hat{y}_i = c_0 + c_1 t_i.$$

The [ ] for the i-th data point is

$$r_i = y_i - \hat{y}_i.$$

Linear regression chooses c_0 and c_1 to make

$$r_1^2 + \cdots + r_m^2$$

as small as possible.
Linear regression is the same closest-reachable-output problem as Unit 2 target revisited (U4-LO4, U4-LO5), but the columns of the matrix now represent model features.

Activity: A line model as a matrix product (U4-LO4)

Consider the data points

$$(0, 1), \qquad (1, 2), \qquad (2, 2),$$

and the line model

$$\hat{y} = c_0 + c_1 t.$$

1. Write the three predicted values in terms of c_0 and c_1.
2. Define

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

   Build a matrix X such that $X\mathbf{c}$ is the vector of predicted values.
3. What does the first column of X represent?

Activity: A line model as a matrix product (U4-LO4) (continued)

4. What does the second column of X represent?

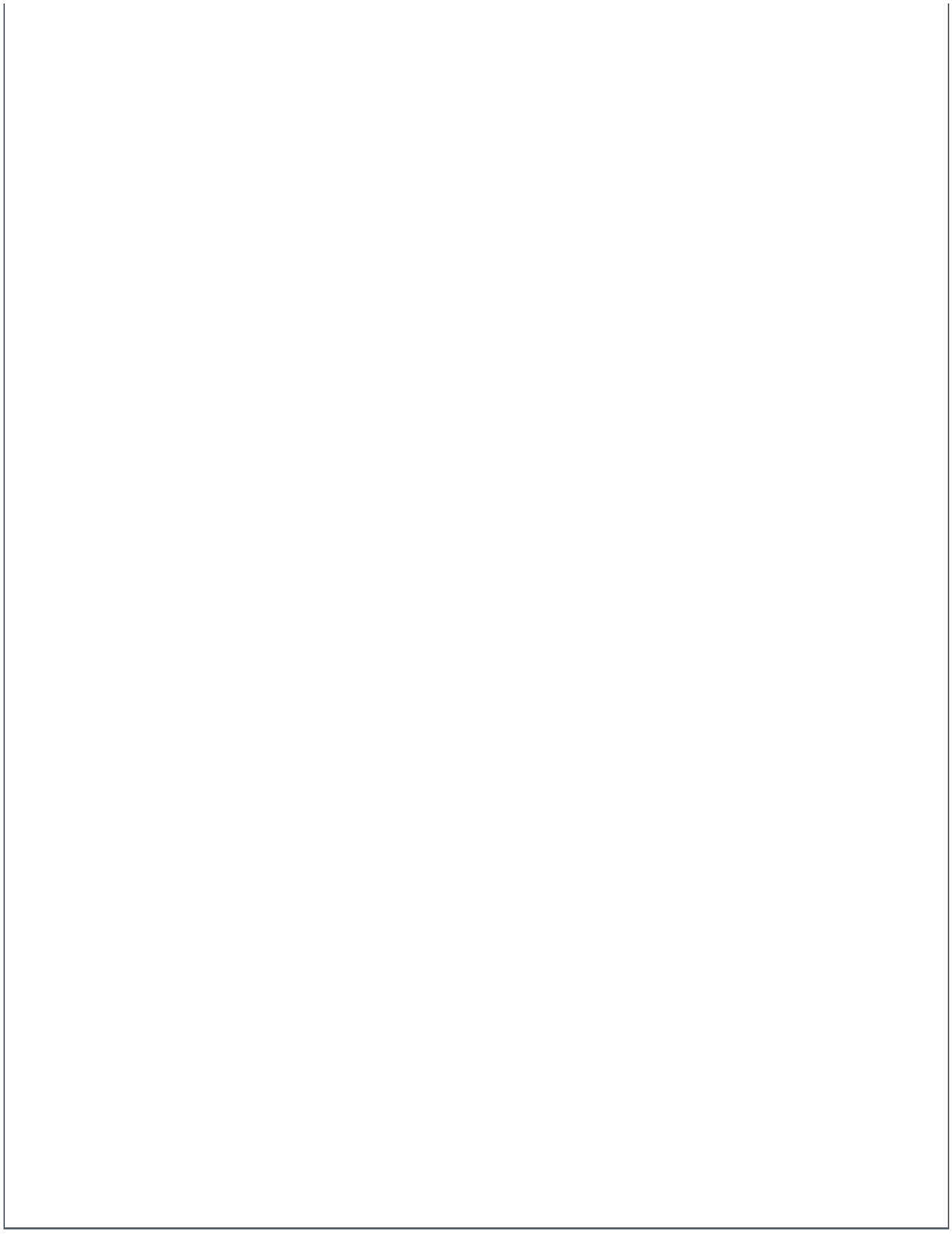
5. Write the residual vector

$$\mathbf{r} = \mathbf{y} - X\mathbf{c}.$$

6. Explain why

$$\|\mathbf{r}\|^2 = \sum_{i=1}^{3} (y_i - \hat{y}_i)^2$$

is the sum of the squared vertical errors in the data plot.



See the figure for the data points, fitted line, and vertical residuals.

For general data (t_i, y_i), define the design matrix

$$X = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \qquad \mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}.$$

Then $X\mathbf{c}$ is the vector of predicted values. Linear regression is the least-squares problem

$$\min_{\mathbf{c}} \|X\mathbf{c} - \mathbf{y}\|^2.$$

The matrix X is called the [ ]. Each column represents one feature used by the model.

Warning: Two different geometric pictures

The projection occurs in $\mathbb{R}^m$:

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}} = \text{proj}_{\text{col}(X)}(\mathbf{y}).$$

In the usual t-y data plot, the residuals are vertical differences

$$r_i = y_i - \hat{y}_i.$$

They are not generally perpendicular distances from the plotted points to the fitted line.

Activity: Fit a line to three data points (U4-LO4, U4-LO5)

Continue with the design matrix and data vector from A line model as a matrix product (U4-LO4). Recall that the data points are

$$(0, 1), \qquad (1, 2), \qquad (2, 2).$$

1. Use Least squares and the normal equations to form

$$X^T X\hat{\mathbf{c}} = X^T\mathbf{y}.$$

Activity: Fit a line to three data points (U4-LO4, U4-LO5) (continued)

2. Solve for

$$\hat{\mathbf{c}} = \begin{bmatrix} \hat{c}_0 \\ \hat{c}_1 \end{bmatrix}.$$

3. Write the fitted line.

4. Compute the fitted-value vector

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}}$$

   and the residual vector

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}.$$

5. Check that

$$X^T\mathbf{r} = \mathbf{0}.$$

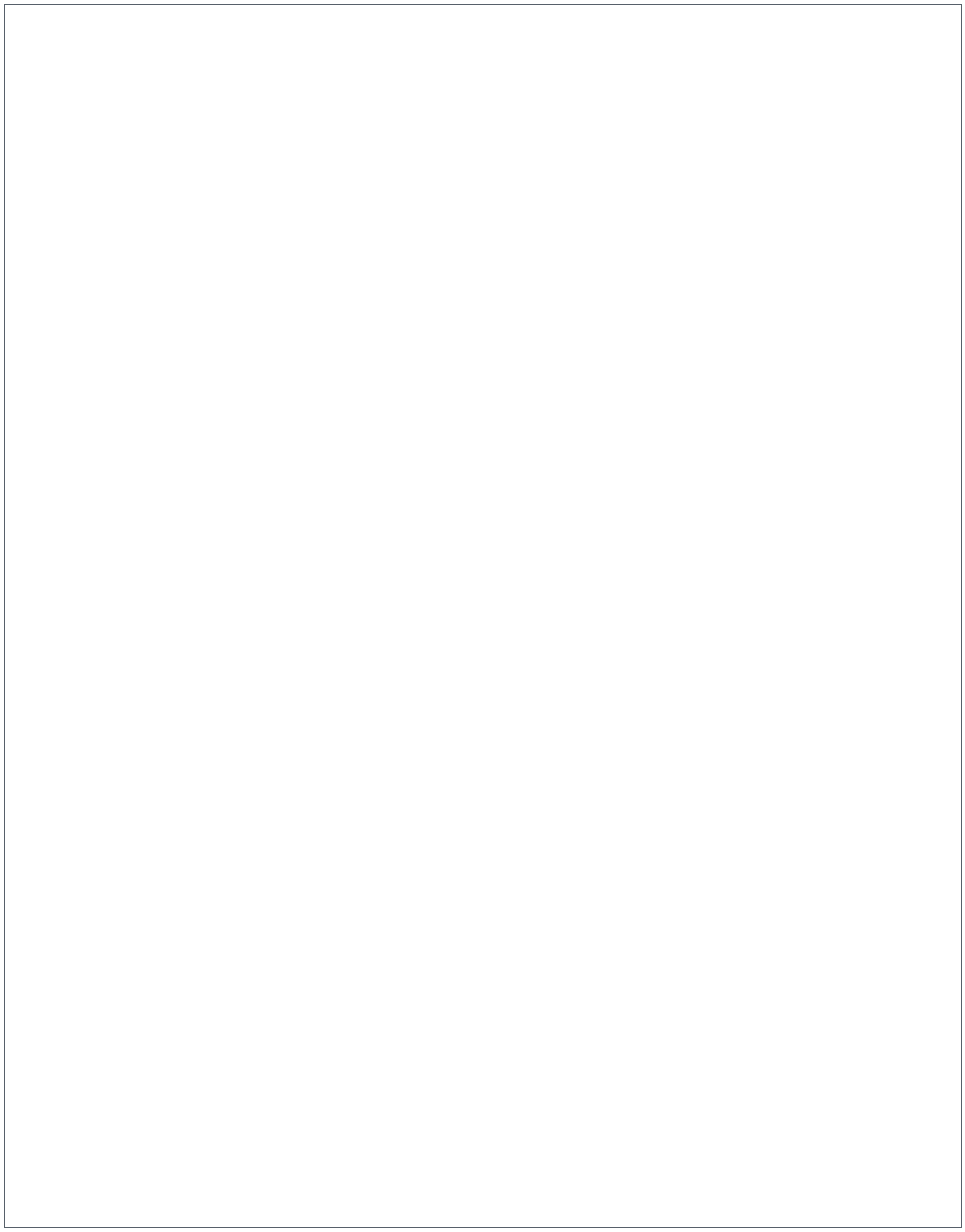
6. Write the two scalar equations contained in $X^T\mathbf{r} = \mathbf{0}$.

7. Compute

$$\|\mathbf{r}\|^2.$$

8. Which object is the fitted vector: $\hat{\mathbf{c}}$ or $\hat{\mathbf{y}}$?

9. In the t-y plot, are the residual segments vertical or perpendicular to the fitted line?



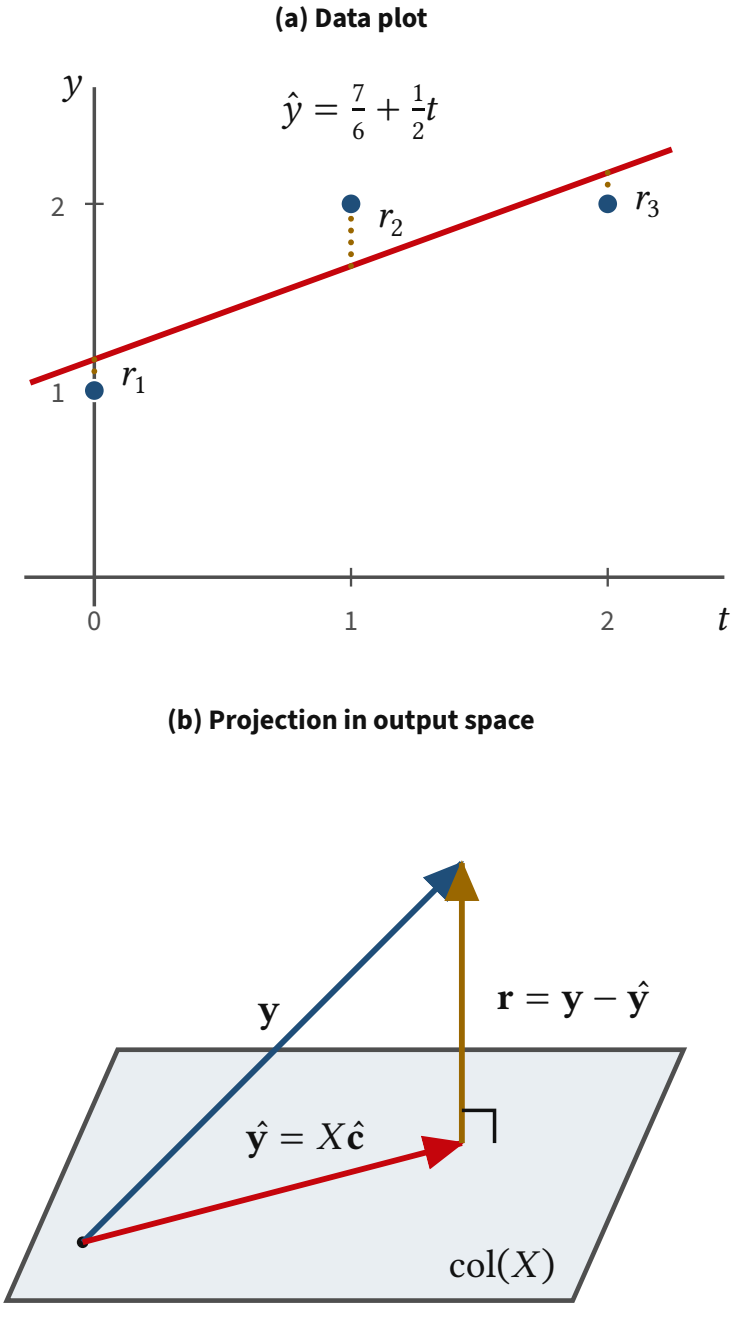


Figure. Two vertically arranged panels show the same linear regression. The upper panel (a) contains three data points, a fitted line, and vertical residual segments. The lower panel (b) shows the data vector y projected onto the column space of X, with a perpendicular residual r.

The two panels in the figure show different spaces: the upper panel (a) is the t-y data plot, while the lower panel (b) is the output space $\mathbb{R}^3$.

Activity: Reading regression code (U4-LO4, U4-LO7)

The following code repeats the line fit from Fit a line to three data points (U4-LO4, U4-LO5).
The NumPy function `np.linalg.lstsq(X, y, rcond=None)` returns a tuple containing the least-squares coefficient vector, an array of residual sums of squares, the rank of X, and its singular values (to which we will come back in Eigenvalues and second-order optimization). We use `[0]` to select the first item, the coefficient vector. The parameter `rcond` controls the relative numerical cutoff used to determine the matrix rank; `rcond=None` uses NumPy's default cutoff.

```
import numpy as np

t = np.array([0.0, 1.0, 2.0])
y = np.array([1.0, 2.0, 2.0])

X = np.column_stack([np.ones_like(t), t])

c = np.linalg.lstsq(X, y, rcond=None)[0]
yhat = X @ c
r = y - yhat

X, c, yhat, r, X.T @ r, np.linalg.norm(r)**2
```

1. What array is produced by `np.ones_like(t)`?
2. What does `np.column_stack([np.ones_like(t), t])` construct?
3. What is the shape of `X`?
4. What does `[0]` select from the value returned by `np.linalg.lstsq`?

Activity: Reading regression code (U4-LO4, U4-LO7) (continued)

5. Why does `c` have two entries?
6. What mathematical objects are stored in `yhat` and `r`?
7. Why is the residual check `X.T @ r` rather than `X @ r`?
8. What should `X.T @ r` be close to?
9. Does a nonzero residual vector mean the code failed?
10. What quantity is computed by `np.linalg.norm(r)**2`?

Explore least squares interactively with the Residual Squares mini-game.